

RAG Scan Stack User Guide and Features

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1 RAG Scan Stack - User Guide & Features Overview

Comprehensive Penetration Testing & Red Team Workflow Platform

1.1 Table of Contents

1. Platform Introduction
 2. Getting Started - Dashboard Overview
 3. Engagement Management
 4. Scope Definition & Intelligence
 5. AI-Powered Scanning
 6. Findings Management
 7. Node & Tunnel Management
 8. Reporting & Export
 9. Advanced Features
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1.2 1. Platform Introduction

1.2.1 What is RAG Scan Stack?

- **Comprehensive Security Testing Platform** for authorized penetration testing and red team operations
- **AI-Enhanced Workflow** with automated scanning, intelligent findings correlation, and multi-agent coordination
- **Centralized Data Collection** from 20+ security tools with unified findings management
- **Advanced Tunneling** with SSH, WireGuard, and C2 framework support for remote assessments

1.2.2 Key Value Propositions

- [SUCCESS] **Workflow Acceleration**: Reduce manual coordination between tools by 80%
 - [SUCCESS] **AI-Driven Intelligence**: Automated reconnaissance, scan recommendations, and finding correlation
 - [SUCCESS] **Unified Findings Management**: Single source of truth for all security findings
 - [SUCCESS] **Remote Assessment Ready**: Built-in tunneling and proxy support for complex network environments
-

1.3 2. Dashboard Overview

1.3.1 Main Navigation Areas

- **** Dashboard****: Real-time status, scan activity, and quick actions
- **[TARGET] Engagements**: Project management and scope tracking
- **[SEARCH] Findings**: Comprehensive vulnerability and findings explorer
- **** Assets****: Network discovery and asset intelligence
- **** AI Agents****: Multi-agent automation and scan orchestration
- **** Nodes****: Remote access and tunnel management
- **** Settings****: Configuration and system management

1.3.2 Dashboard Status Cards

- **System Health**: Service status and performance metrics
- **Active Scans**: Real-time scan progress and queue status

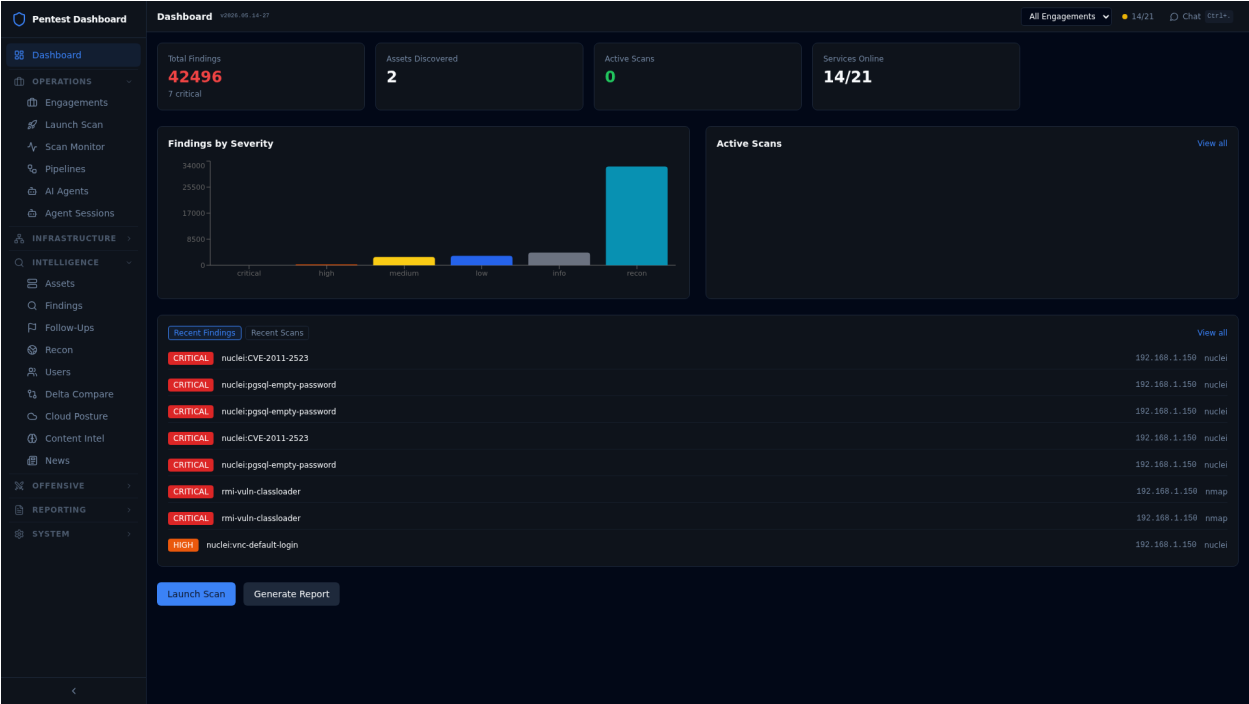


Figure 1: Dashboard Overview

Figure 2: Dashboard Overview

- **Recent Findings:** Latest discoveries with severity indicators
- **Tunnel Status:** Remote node connectivity and proxy availability

1.4 3. Engagement Management Workflow

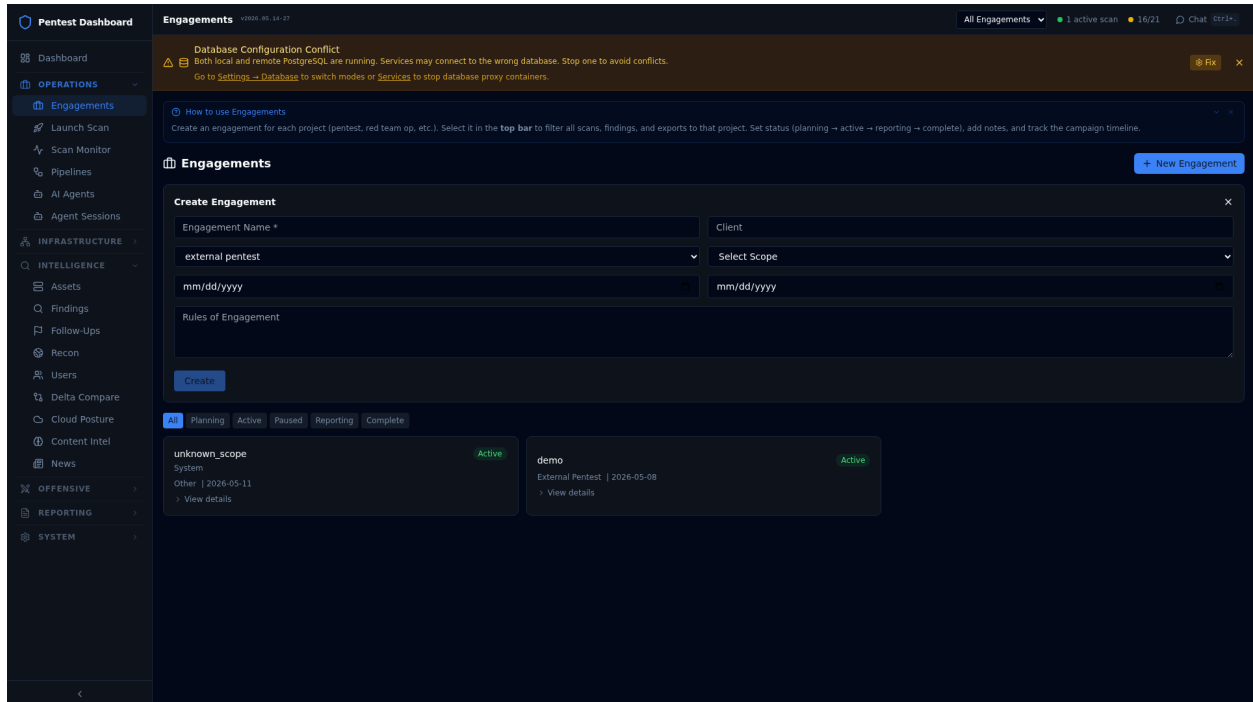


Figure 3: Engagements Management

1.4.1 Step 1: Create New Engagement

1.4.1.1 Navigation

1. Click **Engagements** in main navigation
2. Click **+ New Engagement** button
3. Fill engagement details:

1.4.1.2 Required Fields

Engagement Name: "Acme Corp Assessment 2026"
 Client: "Acme Corporation"
 Start Date: "2026-05-15"
 End Date: "2026-05-30"
 Engagement Type: "Internal Network Assessment"
 Status: "Planning"

1.4.1.3 Optional Configuration

- **Scope Notes:** Internal network ranges and critical systems
- **Rules of Engagement:** Approved testing windows and restrictions
- **Emergency Contacts:** Client SPOC and escalation procedures
- **Testing Methodology:** OWASP, NIST, Custom framework selection

Figure 4: Engagement Management

1.4.2 Step 2: Engagement Timeline & Phases

- **Planning Phase:** Scope definition and approval
- **Active Phase:** Scanning and manual testing
- **Reporting Phase:** Findings compilation and documentation
- **Complete Phase:** Deliverable submission and closure

1.4.3 Step 3: Status Tracking

- **Real-time progress indicators** for each engagement phase
 - **Automated timeline events** from scan completions and findings
 - **Manual milestone tracking** for client meetings and deliverables
-

1.5 4. Scope Definition & Intelligence

1.5.1 Step 1: Define Target Scope

1.5.1.1 Manual Target Entry

Target Types Supported:

- IP Addresses: 192.168.1.0/24, 10.0.0.1-10.0.0.50
- Hostnames: mail.acme.com, *.internal.acme.com
- URLs: https://app.acme.com, http://admin.acme.local
- Domain Names: acme.com, subsidiary.net

1.5.1.2 Bulk Import Options

- **CSV Upload:** Structured target lists with metadata
- **Text Import:** Line-separated targets with auto-detection



Figure 5: Scope Intelligence

Figure 6: Recon Explorer

- **API Integration:** Import from external asset management systems
- **Previous Engagement:** Clone scope from historical assessments

1.5.2 Step 2: Automated Intelligence Gathering

1.5.2.1 OSINT Collection

- **Domain Intelligence:** DNS records, subdomains, certificate transparency
- **Cloud Asset Discovery:** AWS, Azure, GCP resource enumeration
- **Social Engineering Intel:** Employee enumeration, email patterns
- **Technology Stack Detection:** Framework fingerprinting and version detection

1.5.2.2 Passive Reconnaissance

Tools Integrated:

Subfinder - Subdomain enumeration
 DNSx - DNS resolution and validation
 ASNmap - IP range and ownership mapping
 Amass - Comprehensive asset discovery
 Cloudlist - Cloud provider asset enumeration

1.5.3 Step 3: Scope Validation & Approval

- **Asset verification** with client-provided scope documentation
- **Out-of-scope flagging** for discovered assets outside testing boundaries
- **Approval workflow** with client stakeholder sign-off
- **Change management** for scope modifications during testing

1.6 5. AI-Powered Scanning Workflow

1.6.1 Step 1: AI Agent Overview

1.6.1.1 Multi-Agent Architecture

- **** Orchestrator Agent**:** Coordinates scanning strategy and resource allocation
- **[SEARCH] Recon Agent:** Automated reconnaissance and target discovery
- **** Scanning Agent**:** Vulnerability assessment and service enumeration
- **[FAST] Exploit Agent:** Safe exploit validation and proof-of-concept development
- **[ANALYTICS] Analysis Agent:** Finding correlation and risk assessment

1.6.2 Step 2: Automated Scan Pipeline

1.6.2.1 Phase 1: Discovery & Reconnaissance

Scan Sequence:

1. Port Discovery (Masscan/Nmap)
 - + TCP: 1-65535 (adaptive timing)
 - + UDP: Top 1000 ports
2. Service Enumeration
 - + Banner grabbing and fingerprinting
 - + TLS/SSL configuration analysis
 - + HTTP service discovery
3. Technology Detection
 - + Framework identification

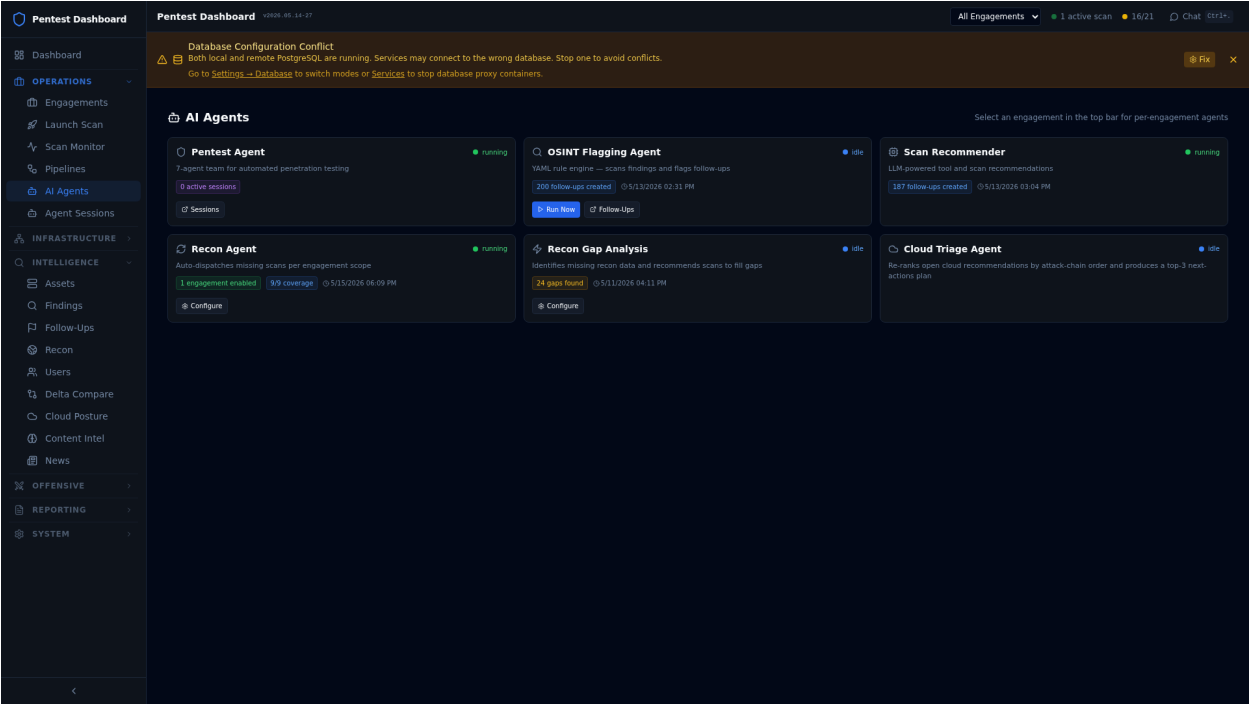


Figure 7: AI Agents Dashboard

Figure 8: AI Agents

- +– Version enumeration
- +– Security header analysis

1.6.2.2 Phase 2: Vulnerability Assessment

Scanning Tools:

Nuclei: 8000+ templates for known vulnerabilities
 ZAP: Dynamic web application scanning
 Playwright: Modern JavaScript application testing
 Custom Scripts: Organization-specific checks

1.6.2.3 Phase 3: Intelligence Correlation

- **Cross-reference findings** across multiple tools
- **Risk scoring** based on exploitability and business impact
- **Attack path mapping** for lateral movement scenarios
- **False positive filtering** using AI-powered validation

1.6.3 Step 3: Manual Scan Initiation

1.6.3.1 Quick Launch Options

- **[LAUNCH] Full Pipeline**: Complete end-to-end assessment
- **[TARGET] Targeted Scan**: Specific service or vulnerability checks
- **** Delta Scan****: Changes since last assessment
- **** Emergency Scan****: Critical vulnerability validation

1.6.3.2 Custom Scan Configuration

Scan Parameters:

- **Timing Template**: [Aggressive, Normal, Polite, Stealth]
 - **Scope Filtering**: [In-scope only, Include adjacent, Full discovery]
 - **Tool Selection**: [Enable/disable specific scanners]
 - **Proxy Settings**: [Direct, Burp Suite, Custom SOCKS]
-

1.7 6. Findings Management

1.7.1 Step 1: Findings Explorer Interface

1.7.1.1 Comprehensive View Options

- **[ANALYTICS] Summary Dashboard**: High-level metrics and trend analysis
- **** Detailed Table****: Sortable, filterable findings with rich metadata
- **** Asset Map****: Network topology with vulnerability overlays
- **** Timeline View****: Chronological discovery and remediation tracking

1.7.2 Step 2: Finding Details & Evidence

1.7.2.1 Rich Finding Information

Finding Attributes:

Title & Description: Clear vulnerability summary
 Severity Rating: Critical, High, Medium, Low, Info
 CVSS Score: Industry-standard risk scoring
 CWE/CVE References: Vulnerability classification
 Affected Assets: Complete host and service details

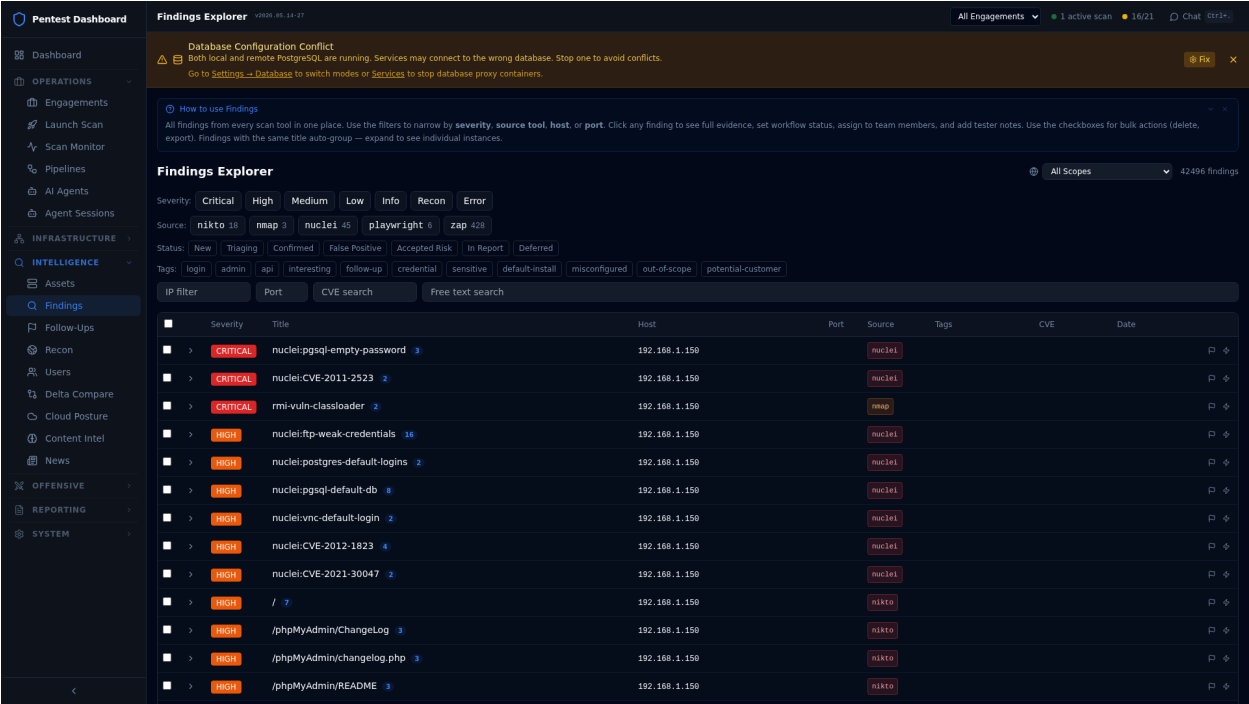


Figure 9: Findings Explorer

Figure 10: Findings Explorer

Discovery Timeline: First seen, last verified dates

1.7.2.2 Evidence Documentation

- **Screenshots**: Automated capture from web scanners
- **[LIST] Raw Output**: Complete tool output and command execution logs
- **[SEARCH] Proof of Concept**: Safe demonstration of vulnerability impact
- **[ANALYTICS] Exploitation Risk**: AI-assessed likelihood and business impact

1.7.3 Step 3: Workflow Management

1.7.3.1 Finding Status Tracking

- **New**: Recently discovered, pending review
- **[SUCCESS] Confirmed**: Validated by security analyst
- **False Positive**: Flagged as invalid finding
- **Retest**: Scheduled for remediation validation
- **[SUCCESS] Fixed**: Confirmed remediation by client

1.7.3.2 Collaboration Features

- **Comments & Notes**: Analyst discussion and client communication
- **Custom Tags**: Organizational categorization and workflow tracking
- **Notifications**: Automated alerts for critical findings
- **External Links**: Integration with ticketing and project management systems

1.8 7. Node & Tunnel Management

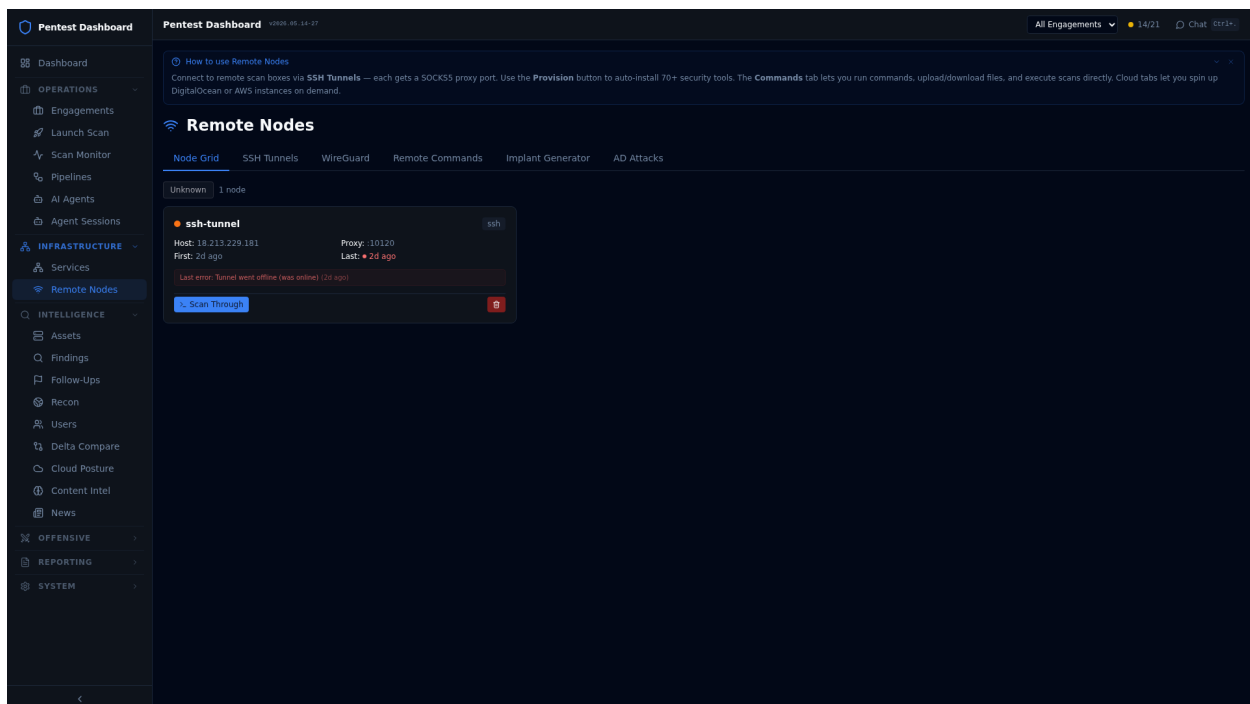


Figure 11: Nodes Overview

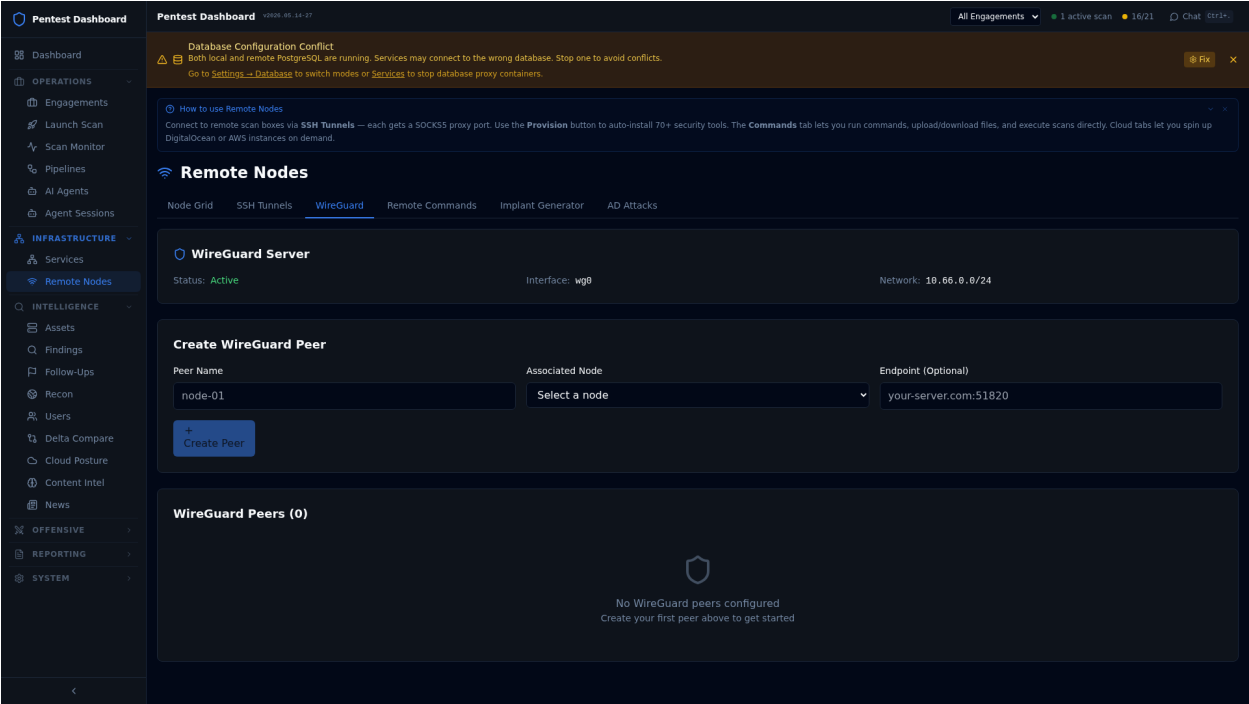


Figure 12: WireGuard Management

Figure 13: Nodes Overview

Figure 14: SSH Tunnels

Figure 15: WireGuard VPN

1.8.1 Step 1: Remote Access Overview

1.8.1.1 Supported Connection Types

- **** SSH Tunnels****: Traditional encrypted tunneling with key-based authentication
- **** WireGuard VPN****: Modern, high-performance VPN tunneling
- **** C2 Frameworks****: Sliver implants for red team operations
- **[NETWORK] HTTP Proxies**: Burp Suite and ZAP integration for web testing

1.8.2 Step 2: WireGuard Management

1.8.2.1 Peer Configuration

WireGuard Setup:

1. Generate Keypair: Automatic server-side key generation
2. IP Allocation: Automatic subnet IP assignment
3. Configuration Export: Copy-paste or QR code for mobile apps
4. Status Monitoring: Real-time connection and traffic statistics

1.8.2.2 Mobile Device Support

- **** QR Code Generation****: Instant configuration for mobile WireGuard apps
- **[LIST] Configuration Export**: Text format for desktop clients
- **** Dynamic Reconfiguration****: Update peers without service restart
- **[ANALYTICS] Usage Statistics**: Bandwidth monitoring and connection analytics

1.8.3 Step 3: SSH Tunnel Operations

1.8.3.1 Tunnel Management

SSH Capabilities:

SOCKS5 Proxy: Dynamic port forwarding for tool traffic
Local Forward: Specific service access (RDP, VNC, etc.)
Reverse Forward: Callback channels for restricted environments
Auto-Reconnect: Resilient connections with health monitoring

1.8.3.2 Security Features

- **** Key-Based Authentication****: No password authentication allowed
- **[SECURE] Certificate Validation**: Host key verification and rotation
- **** Connection Monitoring****: Real-time status and failure alerting
- **[GRAPH] Performance Metrics**: Latency, bandwidth, and stability tracking

1.9 8. Reporting & Export

1.9.1 Step 1: Report Generation Options

1.9.1.1 Standard Report Formats

Export Formats:

Executive Summary: High-level findings for leadership
Technical Report: Detailed findings for IT teams
SARIF: Industry-standard format for security tools
CSV: Spreadsheet analysis and custom processing
JSON: API integration and programmatic access
HAR: HTTP traffic for Burp Suite import

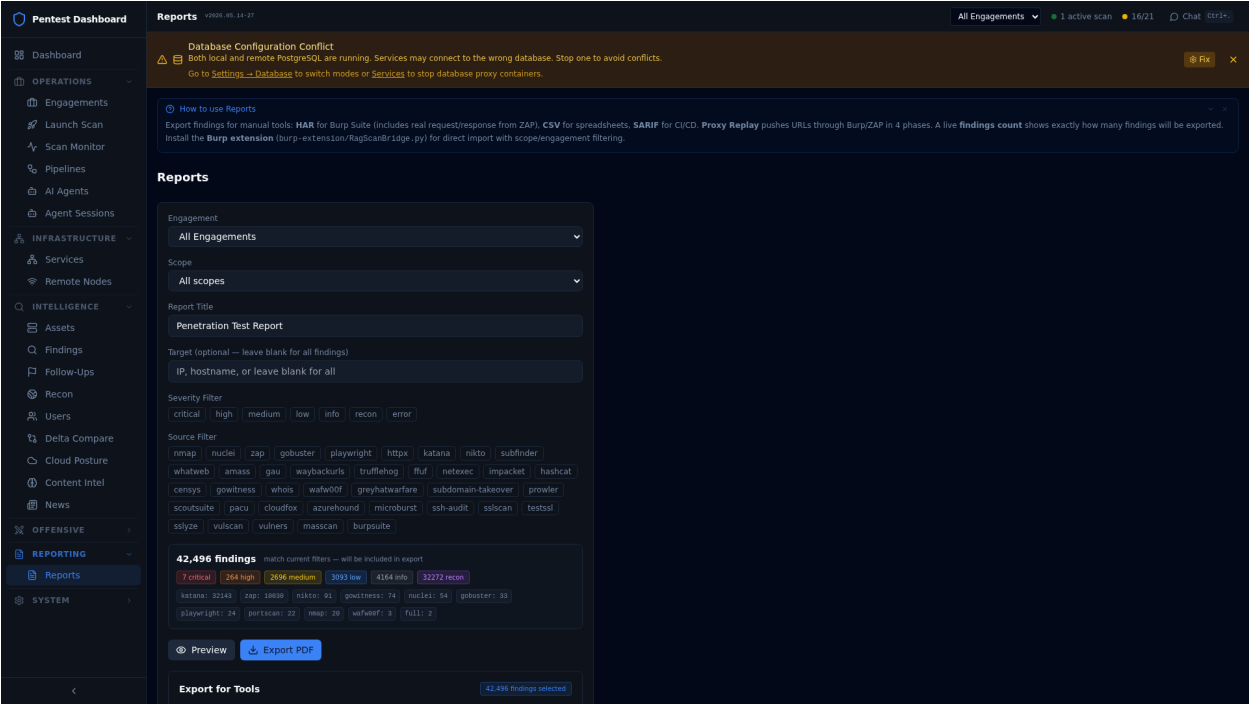


Figure 16: Reports Dashboard

Figure 17: Reports Dashboard

1.9.2 Step 2: Customizable Report Content

1.9.2.1 Report Sections

- **[LIST] Executive Summary:** Business impact and strategic recommendations
- **[TARGET] Scope Overview:** Tested systems and methodology summary
- **[SEARCH] Findings Details:** Complete vulnerability documentation with evidence
- **[ANALYTICS] Risk Analysis:** Prioritized remediation roadmap
- **[GRAPH] Metrics & Trends:** Comparative analysis with previous assessments
- **** Remediation Guidance**:** Step-by-step resolution instructions

1.9.3 Step 3: Integration & Distribution

1.9.3.1 Automated Delivery

- **** Email Distribution**:** Scheduled delivery to stakeholder lists
 - **** API Webhooks**:** Real-time integration with external systems
 - **** Cloud Storage**:** Automatic upload to client document repositories
 - **** Mobile Notifications**:** Critical finding alerts for security teams
-

1.10 9. Advanced Features

1.10.1 Delta Analysis & Trending

- **** Comparative Analysis**:** Before/after remediation validation
- **[GRAPH] Risk Trends:** Improvement tracking over multiple assessments
- **** New Finding Detection**:** Automatic identification of environment changes
- **[ANALYTICS] Remediation Metrics:** Time-to-fix tracking and SLA monitoring

1.10.2 API Security Testing

- **** OpenAPI Import**:** Automatic test generation from Swagger/OpenAPI specs
- **[SEARCH] Endpoint Discovery:** Automated API surface mapping
- **** Authentication Testing**:** Token validation and session management
- **[ANALYTICS] Parameter Fuzzing:** Input validation and injection testing

1.10.3 Cloud Security Posture

- **** Multi-Cloud Support**:** AWS, Azure, GCP configuration assessment
- **** IAM Analysis**:** Permission auditing and privilege escalation detection
- **[PACKAGE] Resource Inventory:** Complete infrastructure mapping and compliance
- **** Misconfiguration Detection**:** Automated security best practice validation

1.10.4 Credential Testing & Management

- **** Secure Vault**:** Encrypted credential storage with audit logging
 - **[TARGET] Targeted Testing:** Username/password validation against discovered services
 - **[ANALYTICS] Success Tracking:** Credential reuse and policy compliance analysis
 - **[SECURE] Integration Security:** Safe handling of client authentication data
-

1.11 Next Steps

1.11.1 For Pentesters & Red Teams

1. **Start with Engagement Creation** - Define project scope and timeline

2. **Configure Target Scope** - Import and validate testing boundaries
3. **Launch AI Scanning Pipeline** - Automated discovery and vulnerability assessment
4. **Review & Validate Findings** - Manual verification and evidence collection
5. **Generate Client Reports** - Professional deliverables with remediation guidance

1.11.2 For Security Managers

1. **Review Dashboard Metrics** - Track team productivity and finding trends
2. **Monitor Service Health** - Ensure platform availability and performance
3. **Analyze Engagement ROI** - Measure testing effectiveness and coverage
4. **Plan Resource Allocation** - Scale infrastructure based on testing demand

1.11.3 Training & Support Resources

- **** Documentation Wiki****: Comprehensive guides at `/about` Documentation
 - **** Video Tutorials****: Step-by-step workflows and advanced techniques
 - **** API Reference****: Complete OpenAPI documentation for automation
 - **** Community Support****: Best practices sharing and troubleshooting assistance
-

1.12 Conclusion

The RAG Scan Stack provides a comprehensive, AI-enhanced platform for modern security testing workflows. By automating routine tasks, centralizing findings management, and providing advanced tunneling capabilities, teams can focus on high-value manual testing while maintaining comprehensive coverage of the attack surface.

Ready to start your first engagement? Begin with the Engagements module and follow this guide for a complete testing workflow.